

Upper-Limb Rehab Game Suite — Design Specification (v0.1 draft)

Camera-tracked upper-limb rehabilitation for inpatient rooms: patient seated on bed/at desk, TV ~10–12 ft away, webcam (MediaPipe) or ZED 2 for tracking. Design grounded in Zhang et al. 2024 (CFI framework) and Cameirão et al. (RGS).

1. Goals

1. Measure each patient's **active range of motion in one plane** and store it as a profile that **every game reads** — all targets appear inside it.
2. Deliver short, adaptive game rounds that hold performance near ~70% success (Yerkes-Dodson / RGS difficulty model).
3. Detect and record **grasp capability** at positions along the ROM.
4. Look and feel **minimal, seamless, hand-drawn** — no intrusive UI during play; big soft shapes readable from 12 ft; calm palette for a 50+ stroke population. Reference: *Melatonin* (pastel gradients, negative space, music-linked feedback).
5. Log everything per round for therapist review and export.

2. Room & hardware

- Patient seated on bed edge or at over-bed desk, facing the TV (10–12 ft).
- Camera at/near the TV, facing the patient (MediaPipe webcam mode), or ZED 2 for true depth. A staff member positions the system; app must include a **setup view** (camera preview + "whole upper body visible" check) that is never shown during patient play.
- All geometry is **body-relative** (origin = shoulder midpoint, unit = shoulder width), so camera distance/height changes between sessions don't invalidate the profile.
- Plane decision (v1): **frontal plane** (what a single RGB camera sees well). Shoulder flexion toward the TV is depth motion → out of scope for MediaPipe; revisit when ZED 2 is in the loop.

3. Tracking

	MediaPipe (v1)	ZED 2 (v2)
Arm joints	Pose Landmarker (shoulder/elbow/wrist), 2D + rough z	Full 3D skeleton, real depth
Grasp	Hand Landmarker on a zoomed crop around the tracked wrist (full-frame hands at 12 ft are unreliable)	No fingers — still needs the crop trick or a near camera
Risk	Grasp detection at distance is the main R&D risk. Fallback if unreliable: dwelling-and-hold (hold position 1–2 s) replaces grasp in any game that asks for it.	Integration effort

Grasp signal (v1 target): binary open/closed from hand-landmark finger curl, smoothed over ~300 ms. Recorded per event with position along the arc.

4. Assessment (the shared "Clinical" step)

Run per arm, first session and re-run any session (doubles as monitoring, per RGS daily calibration). **Gamified as "Dawn Painter"** — the patient's sweeps paint a sunrise, so assessment feels like play, not testing.

Protocol (one plane, ~3 min): 1. **Setup check** (staff view): body visible, tracking stable. 2. **Rest capture**: hand still 1.5 s → home position. 3. **Arc sweep ×3**: "paint the sky" — sweep the arm up/down through maximal comfortable range. Record min/max angle at the shoulder (flexion/extension in the frontal plane v1; wrist radial/ulnar as a second profile type for desk configuration). 4. **Hold test**: at 3 positions (25% / 60% / 90% of arc), hold 2 s inside a soft circle → stability score per position. 5. **Grasp test**: at the same 3 positions, close the hand on a glowing orb → grasp yes/no per position + close time. 6. **Speed/precision**: 5 quick reaches to points on the arc → reaction time, movement time, endpoint error.

Profile schema (per patient, per arm, per plane)

```
{
  "arm": "right", "plane": "frontal-shoulder", "date": "...",
  "arcMinDeg": -20, "arcMaxDeg": 95,
  "holdStability": [0.9, 0.7, 0.3],
  "grasp": { "available": true, "byPosition": [true, true, false], "closeTimeMs": 640 },
  "meanRT": 0.62, "meanMT": 1.1, "endpointErr": 0.14,
  "params": { "rangeScale": 0.7, "radius": 0.30, "lifetime": 4.2 }
}
```

5. Difficulty (shared across all games)

- **Initial** params from assessment: target zone = `rangeScale × measured arc`; target size from endpoint error; target lifetime from RT + MT.
- **DDA between rounds** (Zhang/Cameirão rule): success >70% → harder, <50% → easier, else hold. When easing, use miss locations: mostly misses near arc extremes → shrink `rangeScale` ; otherwise → bigger/slower targets.
- Rounds ≤ 45 s. Rest prompt between rounds; longer-break suggestion every 4 rounds; optional 1-tap pain/fatigue check between rounds (CFI "state assessment") — a face-scale, no text entry.

6. Design language ("minimal & seamless")

- **No HUD during play.** Score/time appear only between rounds. Progress is diegetic: the world itself fills in (sky paints, constellation grows, village lights up).
- Soft two-tone gradient backgrounds per game; ≤3 hues on screen; large soft-edged hand-drawn shapes (wobbly-line texture, paper grain).
- Patient representation: a single elegant cursor (comet / brush tip / firefly) — **no camera feed, no skeleton** in patient view. Skeleton overlay lives in the staff setup view only.
- Feedback is motion + sound: successes bloom and chime in key; misses simply fade (never harsh); gentle BGM; world pulses subtly with the music.
- Compensatory-movement feedback: the **Posture Coach** — a small hand-drawn skeletal mini-figure docked in every activity; faint when posture is good, glows amber on the offending segment (trunk lean, shoulder hike, trunk

rotation) with a ghost overlay of the correct posture and one short spoken cue; green bloom on correction; events logged and fed to DDA. Full spec in `DESIGN_BRIEF.md` §6.

- **Typography:** big rounded type, between rounds only. All prompts also spoken (auto audio instructions, per RGS).

7. The four games

All games read the same profile; targets always inside the measured arc.

G1 — "Constellations" (connect the dots)

- **Mechanic:** stars appear one at a time along the arc; the patient drags a thread of light from star to star. Completing the figure reveals a hand-drawn animal that animates once, then joins the night-sky collection.
- **Trains:** guided reaching, path smoothness, sustained attention.
- **Difficulty knobs:** dot spacing (arc coverage), dot size, required path corridor width, dots per figure.
- **Metrics:** path efficiency (actual/ideal length), smoothness (jerk), time per segment, arc coverage.
- **Reward:** collected constellation creatures (persistent gallery = Zhang's consumption loop).

G2 — "Drift" (the RGS game, reskinned)

- **Mechanic:** soft orbs drift slowly toward the patient's side of the screen along lanes spanning the arc; intercept before they cross the shore line. Three graded stages, unlocked in order (RGS Hitting → Grasping → Placing):
- **Touch** the orb (10 pts).
- Touch **and grasp** it (20 pts).
- Grasp, **carry to the matching-colored tide pool, release** (30 pts).
- **Difficulty knobs (the RGS trio):** orb speed, spawn interval, dispersion across the arc.
- **Metrics:** hit rate, interception point, grasp success/latency, carry path error.

G3 — "Ember Watch" (the Zhang game, reskinned)

- **Mechanic:** night village below; glowing embers drift down toward the rooftops at positions sampled inside the arc. Move the hand to an ember and hold briefly to turn it into a harmless firefly. Round = 30 s (Zhang).
- **Aim-and-commit** = Zhang's shoot mechanic without violence; grasp (if available) can replace hold as the "catch" action.
- **Difficulty knobs (Zhang's):** appearance range (= rangeScale), ember size (strength/precision analog), fall speed.
- **Reward cycle:** points buy lantern styles and village decorations; the village visibly grows/lights over sessions (Zhang's unlock-and-spend loop).

G4 — "Lantern Release" (dedicated reach-and-grasp)

- **Mechanic:** a lantern rests at a spawn point on the arc; reach, grasp, carry along the arc to the release ring near the arc's far extreme, open the hand → lantern floats up and joins a persistent sky of past lanterns.
- **Trains:** transport + timed release (the hardest RGS stage, isolated), grasp-at-extremes (the user's hold/grasp assessment idea as gameplay).
- **Difficulty knobs:** carry distance along arc, release-ring size, lantern "weight" (max carry speed before it slips).
- **Metrics:** grasp position (how far along arc), carry time, drop count, release accuracy.

G5 — "Arc Pong" (pong)

- **Mechanic:** classic pong, minimal: a soft glowing ball floats slowly across a gradient field; the patient's paddle rides **along their measured arc** (shoulder flexion/extension moves it up/down in the frontal plane, or wrist radial/ulnar in the desk plane). Opponent is a gentle AI paddle — or the **therapist/family member on a keyboard or second tracked arm** (two-player = the social element from Zhang's reward-cycle list).
- **Rally-based, not score-punishing:** the sound and background bloom a bit more with every consecutive return; a lost ball just fades and re-serves. Between-round summary shows longest rally.
- **Difficulty knobs:** ball speed, paddle length, serve dispersion across the arc (AI returns are aimed inside `rangeScale × arc` — the DDA decides how much of the arc each rally demands).
- **Metrics:** return rate, rally length, interception point distribution across the arc (how much of the arc is actually being used), paddle smoothness.
- **Why it earns a slot:** continuous tracking (paddle never leaves the arc) trains sustained control rather than discrete reaches — none of G1–G4 covers that; it's also instantly familiar to older patients.
- **Extra skills** (advanced shots layered on top of basic paddle play; therapist-toggled, each its own DDA knob, unlocked as the patient progresses — basic interception is never gated on them):
- **Grip shot (grasp + release):** closing the hand at ball contact catches the ball (brief slow-motion hold); opening it releases a faster, spinning return. Trains timed grasp → hold → timed release under game pressure. Fallback without hand tracking: dwell (hold paddle on the ball for a beat).
- **Spin shot (forearm rotation):** palm orientation at contact sets the spin type — supinated (palm rotating up/front) = topspin, ball dips; pronated (palm rotating down/back) = backspin, ball floats. Adds pronation/supination — a Fugl-Meyer movement no other game in the suite trains. Fallback: coarse palm estimate from pose-model thumb/index/pinky landmarks.
- Skills are also **choice** (Zhang's element): a skilled patient picks which shot to play, not just whether to reach the ball.
- **Bimanual modes:**
- **Split-arc:** one paddle per arm, each covering half the field; AI alternates sides so both arms stay engaged.
- **Coupled (bilateral):** both arms drive one paddle positioned at their average — the affected arm must match the healthy arm's movement.
- **Self-rally:** affected arm vs healthy arm on opposite sides; built-in same-session inter-arm comparison for the data log.
- Requires a per-arm profile for both arms; each arm's returns stay inside its own measured arc.
- **Multiplayer modes:**
- **Vs therapist/family (same camera):** MediaPipe tracks 2 people (numPoses: 2); the second person's arm drives the opposing paddle. The companion needs no assessment — their paddle uses a default arc; only the patient's side is ROM-bounded and DDA-controlled.
- **Vs companion (keyboard/remote):** simplest fallback — opponent paddle on arrow keys or a phone browser page acting as a controller.
- **Patient vs patient (stretch goal):** two beds/rooms over local network; each patient plays inside their own assessed arc — asymmetric arcs are fine because each side's returns are aimed within the receiver's range. High motivational value (Zhang's social element), but network + privacy review needed → later milestone.

Session flow: pick arm → (Dawn Painter if due) → therapist or patient picks a game → rounds with DDA until stop rule (fatigue check / time / patient exit).

8. Data & therapist view

- Per round: game, params, hit rate, kinematic metrics above, grasp events.
- Trends screen: arc range over sessions, % arc used, smoothness, grasp map.

- Export JSON (v1); CSV later. All local; no PHI beyond a chosen ID string.

9. Milestones

1. **M1 — Platform + plane:** one-plane arc assessment (Dawn Painter), profile storage, staff setup view, design-language shell. ✅ partially built (v0 prototype exists; needs arc-angle model + new aesthetic).
2. **M2 — Constellations + Arc Pong** (no grasp dependency; safest first games — one discrete-reach, one continuous-control).
3. **M3 — Grasp R&D:** wrist-crop hand tracking at distance; fallback dwell.
4. **M4 — Drift + Lantern Release** (grasp-dependent stages gated on M3).
5. **M5 — Ember Watch + reward economy across games.**
6. **M6 — ZED 2 backend; second plane (toward-TV flexion).**

Open questions: exact joint-angle definition per plane (goniometry-matched?), pain-scale wording per clinical team, patient ID scheme, bilateral/mirror modes later? Since Pong's spin shot uses pronation/supination, should the assessment add a "rotate palm up/down" step at the grasp-test positions to record available rotation range?